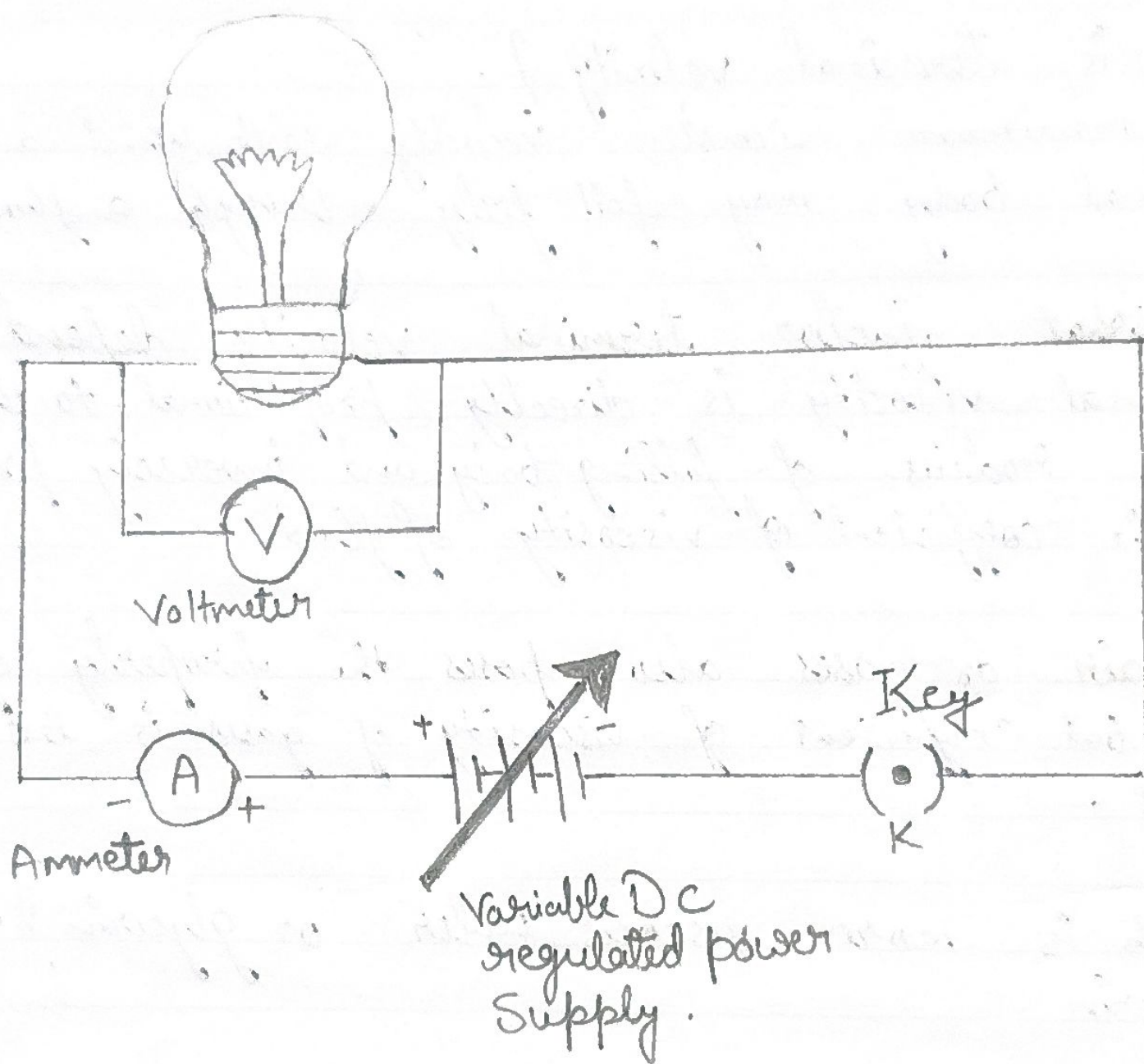




Δ Figure

Experiment No - 2

Object - To verify Stefan's Law by electrical Method.

Apparatus Required -

1. Continuous Variable DC Regulated Power Supply 0-20 Volts.
2. D.C ammeter
3. D.C voltmeter
4. Bulb
5. Connection Leads

Formula Used -

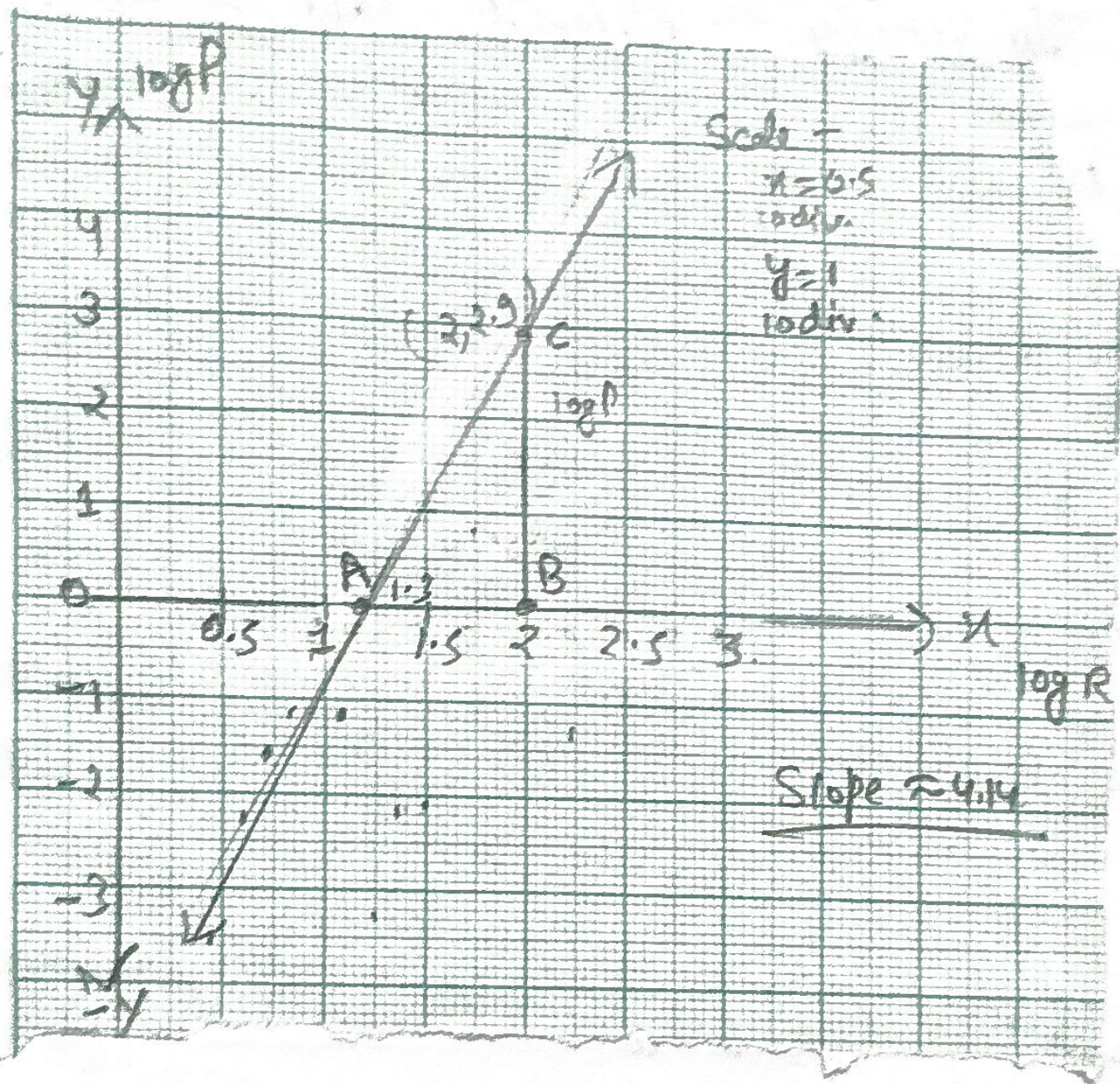
$$\log_{10} P = \alpha \log_{10} T + \log_{10} C.$$

Where,  $P$  = Total power emitted by a body at temperature  $T$ ,

$\alpha$  = power of  $T$  close to 4

$T$  = temperature of a body

$C$  = Some constant depending on the material and area of such a body.



Principle —

Acc. to Stefan's law power radiated from a black body is proportional to the fourth power of its absolute temperature.

$$\text{i.e. } P \propto T^4$$

$$\text{or } \log P \propto 4 \log T \quad \text{--- (1)}$$

Resistance of the tungsten filament of the electric bulb is —

$$R \propto T$$

$$\text{or } \log R \propto \log T \quad \text{--- (2)}$$

From eq<sup>n</sup> (1) & (2)

$$\log P / \log R = 4.$$

Observation & Tabulations

| S.No | Filament Voltage | Filament Current | Filament Resistance | Power Radiated<br>$P = VI$ | $\log R$ | $\log P$ |
|------|------------------|------------------|---------------------|----------------------------|----------|----------|
| 1.   | 0.03             | 0.01             | 3                   | $3 \times 10^{-4}$         | 0.4771   | -3.5229  |
| 2.   | 0.06             | 0.02             | 3                   | $1.2 \times 10^{-3}$       | 0.4771   | -2.9208  |
| 3.   | 0.10             | 0.029            | 3.44                | $2.9 \times 10^{-3}$       | 0.5366   | -2.5376  |
| 4.   | 0.13             | 0.035            | 3.77                | $4.5 \times 10^{-3}$       | 0.5763   | -2.3468  |
|      |                  |                  |                     | Teacher's Signature _____  |          |          |

## Calculation —

$$\text{Point A} = (x_1, y_1) = [1.30, 0.00]$$

$$\text{Point B} = (x_2, y_2) = [2.00, 0.00]$$

$$\text{Point C} = (x_3, y_3) = [2.00, 2.90]$$

$$\text{Slope } (\alpha) = \tan \theta = \frac{BC}{AB} = \frac{2.9}{0.7} \approx 4.14.$$

|    |      |       |      |                       |        |          |
|----|------|-------|------|-----------------------|--------|----------|
| 5. | 0.19 | 0.043 | 4.41 | $8.1 \times 10^{-3}$  | 0.6444 | - 2.0410 |
| 6. | 0.22 | 0.047 | 4.6  | $10.3 \times 10^{-3}$ | 0.6628 | - 1.9872 |
| 7. | 0.28 | 0.051 | 5.4  | $14.2 \times 10^{-3}$ | 0.7324 | - 1.8477 |
| 8. | 0.30 | 0.053 | 5.6  | $15.9 \times 10^{-3}$ | 0.7482 | - 1.7986 |

### Results -

Slope of the graph  $\approx 4$ , which verifies the Stefan's law.

### Precautions - & Source of Error -

1. Use the bulb having tungsten filament.
2. Increase the current in steps.
3. Note down the voltage reading after every change in current.
4. Choose the variable DC supply in appropriate range.
5. Connection should be tight.
6. Fluctuation in the voltage / current supply.

## Viva - Voice Questions -

Q1 What is Stefan's law?

The amount of thermal radiation emitted per second per unit area of the surface of a black body is directly proportional to the fourth power of the absolute temp.

Q2 What is a perfect black body radiation?

A perfect black body is one which absorbs completely the radiations of all wavelength incidents on it. Absorbance is equal to 1.

Q3 What is black body radiation?

A perfect black body is heated to a high temperature it emits radiation consisting of all possible wavelengths. The radiation thus emitted is called the black body radiation.

Q4 Define emissive power of a body?

Emissive is defined as the ratio of the emissive power of a body to the emissive power of a black body at the same temperature.

Q5 Define emissive power of a body.

Emissive power of a body at a particular temperature is defined as the total energy radiated per second.

Q6 Can a perfect black body cannot be realised in practice?  
A perfect black body cannot be realised in practice.

Q7 What is Thermal radiation?

The energy emitted by body in the form of radiation by virtue of its temperature is called thermal radiation of energy.

Q8 What is the speed with which thermal radiation travels?  
Thermal radiation travels with a speed of light  
i.e.  $3 \times 10^8$  m/s.

Q9 What is the wavelength range of thermal radiation?

The wavelength range of thermal radiation is from  $8 \times 10^{-7}$  m to  $3 \times 10^{-4}$  m.

Q10 Application —

It is used to measure the temperature of hot objects without direct contact, such as in furnaces and industrial processes.